

Multiple Choice

1. **Answer:** A

Options: A) Temperature, humidity, and air currents; B) Plant species, root depth, and wind speed; C) Soil fertility, irrigation methods, and cloud cover; D) Water salinity, leaf color, and time of day; E) Leaf surface area, stem length, and moon phase

Note: Correct option: A - Temperature, humidity, and air currents

2. **Answer:** B

Options: A) Frame shift mutations and nucleotide repeats; B) Base substitutions and small insertions and deletions; C) Base additions and large insertions; D) Silent mutations and missense mutations; E) Duplication mutations and triplet repeat expansions

Note: Correct option: B - Base substitutions and small insertions and deletions

3. **Answer:** C

Options: A) It is created by the direct action of natural selection.; B) It tends to be reduced by the processes involved when diploid organisms produce gametes.; C) It must be present in a population before natural selection can act upon the population.; D) It is always increased by the processes involved when diploid organisms produce gametes.; E) It is not affected by the environmental changes.

Note: Correct option: C - It must be present in a population before natural selection can act upon the population.

4. **Answer:** C

Options: A) Aristotle, Darwin, and Lamarck; B) Aristotle, Lyell, and Darwin; C) Aristotle, Linnaeus, and Cuvier; D) Aristotle, Cuvier, and Lamarck; E) Linnaeus, Darwin, and Lamarck

Note: Correct option: C - Aristotle, Linnaeus, and Cuvier

5. **Answer:** A

Options: A) Earthworms use gills, frogs use both gills and lungs, fish use lungs, and humans use skin.; B) Earthworms and humans use lungs, frogs and fish use gills.; C) All use gills for obtaining oxygen.; D) Earthworms and humans use gills, frogs use skin, and fish use lungs.; E) Earthworms use lungs, frogs use gills, fish use skin, and humans use gills.

Note: Correct option: A - Earthworms use gills, frogs use both gills and lungs, fish use lungs, and humans use skin.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: This Google-based search was useful to identify an appropriate diagnosis in complex immunological and allergic cases. Computing skills may help to get better results.

7. **Answer:** False

Options: A) True; B) False

Note: Numerous studies have found that taxation is one of the most effective policy instruments for tobacco control. However, these findings come from countries that have market economies where market forces determine prices and influence how cigarette taxes are passed to the consumers in retail prices. China's tobacco industry is not a market economy; therefore, non-market forces and the current Chinese tobacco monopoly system determine cigarette prices. The result is that tax increases do not necessarily get passed on to the retail price.

8. **Answer:** True

Options: A) True; B) False

Note: Head and neck-specific QOL measures are necessary and should include domains that reflect ES, SC, and AP.

9. **Answer:** True

Options: A) True; B) False

Note: Findings suggest that traffic law reforms in order to have an effect on both traffic fatality and injury rates reduction require changes in police enforcement practices. Last, this case also illustrates how the diffusion of successful road safety practices globally promoted by WHO and World Bank can be an important influence for enhancing national road safety practices.

10. **Answer:** True

Options: A) True; B) False

Note: In the present study, FT and androstenedione were statistically significantly correlated with sexual desire in the total cohort of women. ADT-G did not correlate more strongly than circulating androgens with sexual desire and is therefore not superior to measuring circulating androgens by mass spectrometry.

Long Form Response

11. **Answer:** Radioactive isotopes can be used to change the color of cells, making them easier to observe under a microscope.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** It was the incomplete combustion products of the kerosene, identified as ethylene, that ripened the fruits.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key